Jiahao Cao

Curriculum Vitae | Email: jiahao.cao@uth.tmc.edu

Education

Renmin University of China 2019/9 – 2024/6
Ph.D. in Statistics. *Beijing, China*

Chinese University of Hong Kong, Shenzhen 2022/9 – 2023/2
Visiting Ph.D. student. *Guangdong, China*

University of California, Berkeley 2019/1 – 2019/5
Visiting student *California, USA*

Nankai University 2015/9 – 2019/6
B.S. in Statistics *Tianjin, China*

Research Experience

University of Texas Health Science Center at Houston 2024/10 – Now
Postdoctoral Research Fellow *Houston, USA*

Current Research Interests

My research focuses on developing rigorous, interpretable, and computationally scalable statistical methods for complex structured data, with a particular emphasis on Bayesian inference, uncertainty quantification, and modern statistical machine learning. I aim to integrate methodological innovation with real-world applications in public health, environmental science, and social systems, especially where data are spatially indexed, temporally evolving, high-dimensional, or subject to measurement and privacy-related uncertainties.

Methodological Interests

- **Bayesian and nonparametric modeling**, including Bayesian additive regression trees, mixture models, hierarchical models, Bayesian networks, and scalable posterior computation
- **Spatial and spatio-temporal statistics**, including Gaussian processes, log-Gaussian Cox processes, small-area estimation, spatially varying coefficient models, disease mapping, and network-based spatial models
- **Complex structured data analysis**, including compositional data, functional data, point process data, high-dimensional regression, and constrained or shape-preserving inference
- **Dynamic modeling and latent structure discovery**, including clustering, community detection, nonlinear time series, change point detection
- **Robust and privacy-aware statistical inference**, including measurement error, data suppression, spatial misalignment, and uncertainty propagation in public-use datasets
- **Modern AI and uncertainty quantification**, including interpretable statistical learning, Bayesian machine learning, and uncertainty-aware deep learning for scientific and public health applications

Application Areas

- **Public health and epidemiology**, including cancer surveillance, opioid overdose mortality, health disparities, and social determinants of health
- **Environmental and spatial health studies**, including exposure assessment, disease risk mapping, and integration of geospatial, registry, administrative, and survey data
- **Interdisciplinary applications** in urban systems, sports analytics, and financial/economic functional time series, where spatial, temporal, or functional structures are central

Publication

Published:

- **Jiahao Cao**, Kehe Zhang, Guanyu Hu, Kelly Nelson, Cici Bauer (2026). Bayesian Spatio-temporal Small Area Modeling: A Case Study Investigating the Late-Stage Melanoma Incidence in Texas. *Journal of the Royal Statistical Society: Series C*
- Jack Cordes, Cici Bauer, Dana Bernson, Marc R Larochelle, Wenjun Li, **Jiahao Cao**, Olaf Dammann, Thomas J Stopka (2026). The All-Payer Claims Database After the Gobeille Supreme Court Decision: Assessing the Impact on Fatal Opioid-Related Overdose Research. *Health Services Research*
- **Jiahao Cao**, Shiyuan He, Bohai Zhang (2023). Spatial Linear Regression with Covariate Measurement Errors: Inference and Scalable Computation in a Functional Modeling Approach. *Journal of Computational and Graphical Statistics*

Under Review:

- **Jiahao Cao**, Shiyuan He, Bohai Zhang (2024+). Functional BART with Shape Priors: A Bayesian Tree Approach to Constrained Functional Regression. Major revision in *Bayesian Analysis*
- **Jiahao Cao**, Hou-Cheng Yang, Guanyu Hu (2026+). How do professional players select their shot locations? An analysis of Field Goal Attempts via Bayesian Additive Regression Trees. submitted to *Journal of Computational and Graphical Statistics*
- **Jiahao Cao**, Qingpo Cai, Lance A. Waller, DeMarc A. Hickson, Guanyu Hu, Jian Kang (2026+). What Influences the Field Goal Attempts of Professional Players? Analysis of Basketball Shot Charts via Log Gaussian Cox Processes with Spatially Varying Coefficients. submitted to *Statistical Analysis and Data Mining: An ASA Data Science Journal*

Working Papers:

- **Jiahao Cao**, Kehe Zhang, Cici Bauer (2026+). Bayesian ACCESS for Understanding Latent Epidemic Trajectories from Publicly Released Suppressed Data: Application to U.S. Opioid-related Overdose Mortality. to be submitted to *Journal of the American Statistical Association*
- **Jiahao Cao**, Eoghan O Neill, Maria Grith, Anastasija Tetereva, and Guanyu Hu. Nonlinear Autoregressive Models for Functional Time Series with Bayesian Additive Regression Trees. to be submitted to *Journal of Business and Economic Statistics*
- Kehe Zhang, **Jiahao Cao**, Guanyu Hu, Cici Bauer (2026+). Bayesian Heterogeneity Learning for Spatial-Temporal Count Data: An Application to Melanoma Cancer Registry Data in Texas.
- **Jiahao Cao**, Alexander Testa, Nicholas Reger, Qian Xiao, Cici Bauer. Nighttime Illumination and Crime Dynamics in Houston: A Longitudinal Cross-Lagged Analysis.
- Bayesian Regression Tree and Forest with High-dimensional Compositional Predictors
- Bayesian Dynamic Community Detection

Awards & Honors

Outstanding Graduates
Renmin University of China *2024*

Winner (1st) of the sub-competition: parameter estimation
KAUST Competition on Spatial Statistics for Large Datasets [link] *2021*

First-class academic scholarship
Renmin University of China *2019, 2020, 2022*

First prize in Tianjin
China Undergraduate Mathematical Contest in Modeling *2018*

"Gong Neng" scholarship
Nankai University *2016,2017*

Teaching Experience

Teaching Assistant
Institute of Statistics and Big Data, Renmin University of China *Beijing, China*

- Fall 2023: Computer Skills for Data Science
- Fall 2022: Computer Skills for Data Science
- Fall 2021: Data Science and Artificial Intelligence Algorithms
- Spring 2021: Bayesian Modeling and Inference
- Fall 2020: Computer Skills for Data Science

Academic Service

Reviewed for Journal: Biometrics, Annals of Applied Statistics, Journal of Biopharmaceutical Statistics, Applied Stochastic Models in Business and Industry

Conference Presentations

2025 ICSA Applied Statistics Symposium
Invited speaker: Recent Advances in the Modeling and Computations for Complex Spatial and Functional Data 2025

EAC-ISBA Conference
Invited speaker: Applied nonparametric Bayesian advances in and beyond biomedical research 2024

Workshop on Spatial Statistics at Renmin University of China
Speaker 2024

Joint Conference on Statistics and Data Science in China (JCSDS)
Speaker in a Contributed Session 2023

Skills

Programming Tools and Languages: R; Python; C++ (Rcpp)